



Using Neighborhood Data in Aging Research

The National Neighborhood Data Archive (NaNDA)

GSA Annual Scientific Meeting Online Workshop Series 2026

Wednesday, October 7, 2026, 12:00 to 2:00 PM ET

Funding & support



- National Institutes of Health
 - National Institute on Aging, R01AG109115
 - National Institute of Nursing Research, U01NR020556
 - National Institute on Minority Health and Health Disparities, U01NR020556



National Institute
on Aging



National Institute
of Nursing Research



National Institute
on Minority Health
and Health Disparities

Your presenters

- **Philippa Clarke** she|her
 - Research Professor, NaNDA Co-Principal Investigator
- **Grace Noppert** she|her
 - Research Associate Professor, NaNDA Co-Principal Investigator
- **Lindsay Gypin** she|her
 - Data Librarian and Senior Project Manager
- **Robert Melendez** he|him
 - Data Analyst
- **Will Clary** he|him
 - Research Data Analyst



Today's plan

Time (ET)	What we are doing	Who
12:00	Welcome and logistics	Philippa & Lindsay
12:05	What is NaNDA? Overview and repository tour	Lindsay & Philippa
12:30	NaNDA in action: examples from aging research	Philippa & Grace
12:55	Choose your breakout	Lindsay
1:00	Breakouts: beginner and intermediate	All
1:45	Reconvene and report back	Grace
1:55	Wrap-up and resources	Lindsay

Housekeeping

- Captions are available. Turn them on with Show Captions in your Zoom toolbar.
- Stay muted in the full group. You are welcome to unmute in your breakout.
- Chat is for comments and links. Q&A is for questions you want us to answer.
- GSA Digital Programs is running the technical side. Message the host if something breaks.
- You do not need to install anything. Everything today runs in a browser.

Workshop materials

EVERYONE

Presentation slides

Slides from all of today's presentations.

BEGINNER

Participant Handout

For Choosing Your First NaNDA Dataset: your worksheet, plus the breakout's slide content. Scavenger hunt answers are at the bottom.

INTERMEDIATE

Practice data and code

For Linking NaNDA with Your Data: practice data and sample merge code in Stata, R, and Python.

Available through the workshop DOI. The link is in the chat.

**In the chat: what do you
wish you could measure
about the neighborhoods?**



By 2:00 you will be able to

- Identify which NaNDA datasets fit your research question, by geographic level, years covered, and topic
- Link NaNDA to your own data using geographic identifiers, including when to use tract_fips10 or tract_fips20, and how ZIP codes differ from ZCTAs
- Frame a research question that combines neighborhood measures with your individual-level aging outcomes

Two options. You pick one at 12:55.

- **Beginner: Choosing Your First NaNDA Dataset**

- Census tracts, ZCTAs, and which matches your data
- Matching NaNDA years to your collection period
- Reading a codebook and finding the identifier
- Hands-on time in the repository

- **Intermediate: Linking NaNDA with Your Data**

- Identifiers and the pitfalls that break a merge
- When you need a crosswalk, and which one
- A live merge you can follow along with
- Drafting a research question that uses both

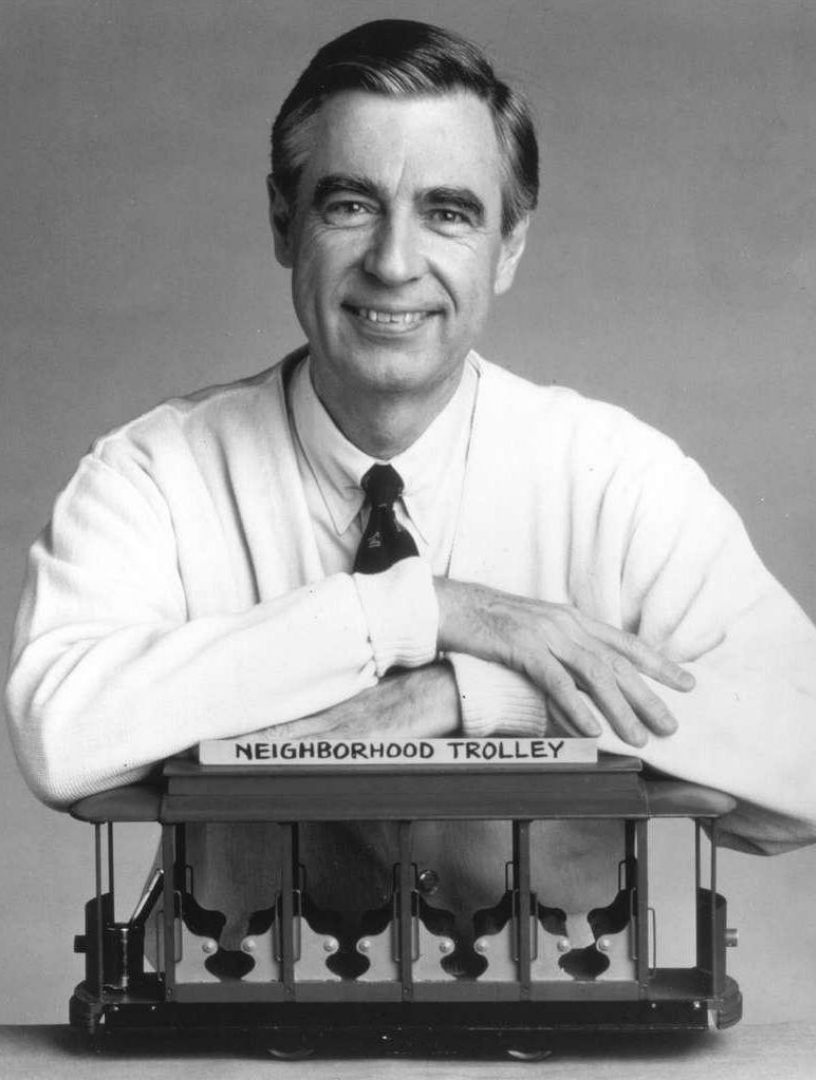
NaNDA overview

An overview of the National Neighborhood Data
Archive, and a tour of the repository



What we'll cover

- 01 Why neighborhood context matters for aging research
- 02 What NaNDA is, and what is in it
- 03 Which NaNDA datasets are already linked to the studies you use
- 04 Geography: the part that trips people up
- 05 **A live tour of the repository**



What is your neighborhood?

A few blocks.

A school catchment.

A census tract, because that is what the data came in.

Your choice changes your findings

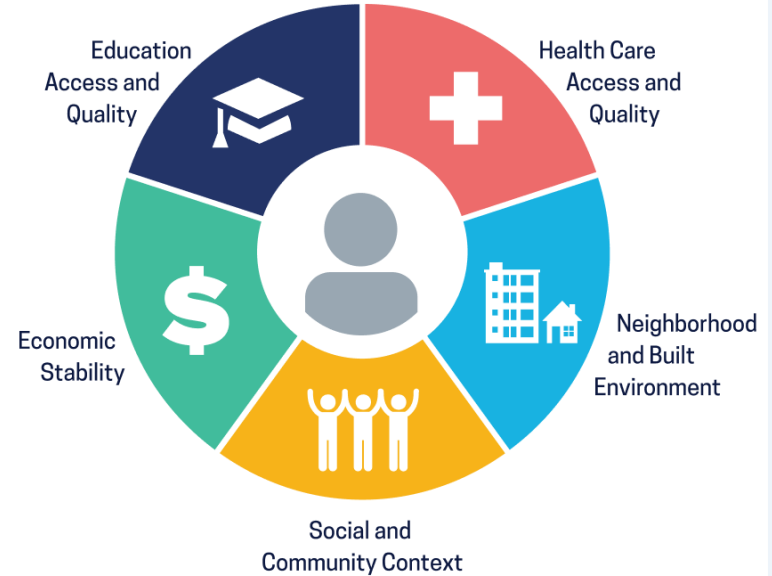
SOCIAL DETERMINANTS OF HEALTH

Neighborhood and built environment is one of the five domains.

It is also the one most researchers have the least data on.

Healthy People 2030, U.S. Department of Health and Human Services.

Social Determinants of Health



Social Determinants of Health
Copyright-free

 Healthy People 2030

What we mean by neighborhood

Built and physical

- Walkability, public transit, traffic
- Food stores, parks and recreation
- Healthcare, hospitals
- Air pollution, climate, air conditioning
- Broadband

Social and economic

- Disadvantage, affluence, gentrification
- Mortgage lending, housing affordability
- School quality, employment, crime
- Voter registration, turnout, and partisanship

Where data come from

1

Ask people

Survey items on safety, cohesion, park quality. Captures perception, but it is your own data collection.

2

Go look

Systematic social observation, in person or by street view. Rich, and expensive.

3

Use data that already exist

Census and ACS, business point data, parcel data, weather stations, administrative records. **This is what NaNDA is built from.**

2019

**Every research team was
building the same
neighborhood measures
from scratch.**

In 2019, NaNDA was founded

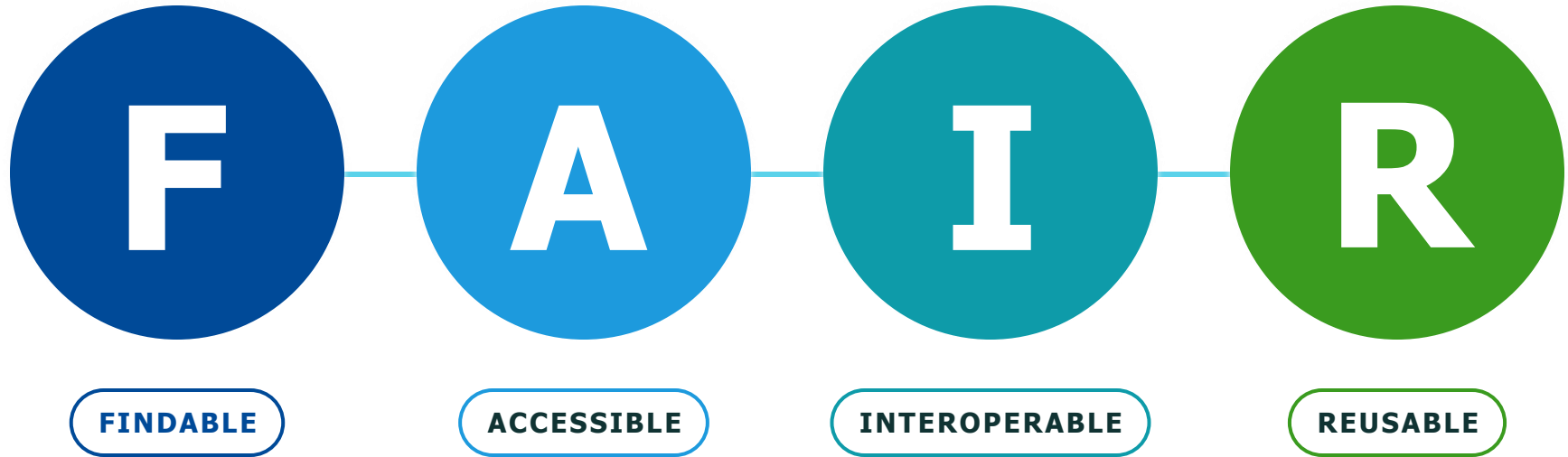


What is NaNDA?

The National Neighborhood Data Archive (NaNDA) is an **open data repository**. It provides **FAIR datasets** about physical and social environment that researchers can easily combine with their own data.



FAIR data principles



Examples of neighborhood topics include:



Health care services

(health care providers, residential care facilities, hospitals)

Parks

(public parks, national and state parks)

Social Services

(crisis centers, shelters, child day care services)

Transportation

(street connectivity, public transit stops, traffic)

Broadband

(internet availability, speed, adoption, and access)

Liquor, tobacco and dispensaries

(counts and density measures)

Crimes

(crime reported by the FBI)

Grocery stores

(counts and density measures)

Data updates

Business Openings & Closings

Quality of Public Transit

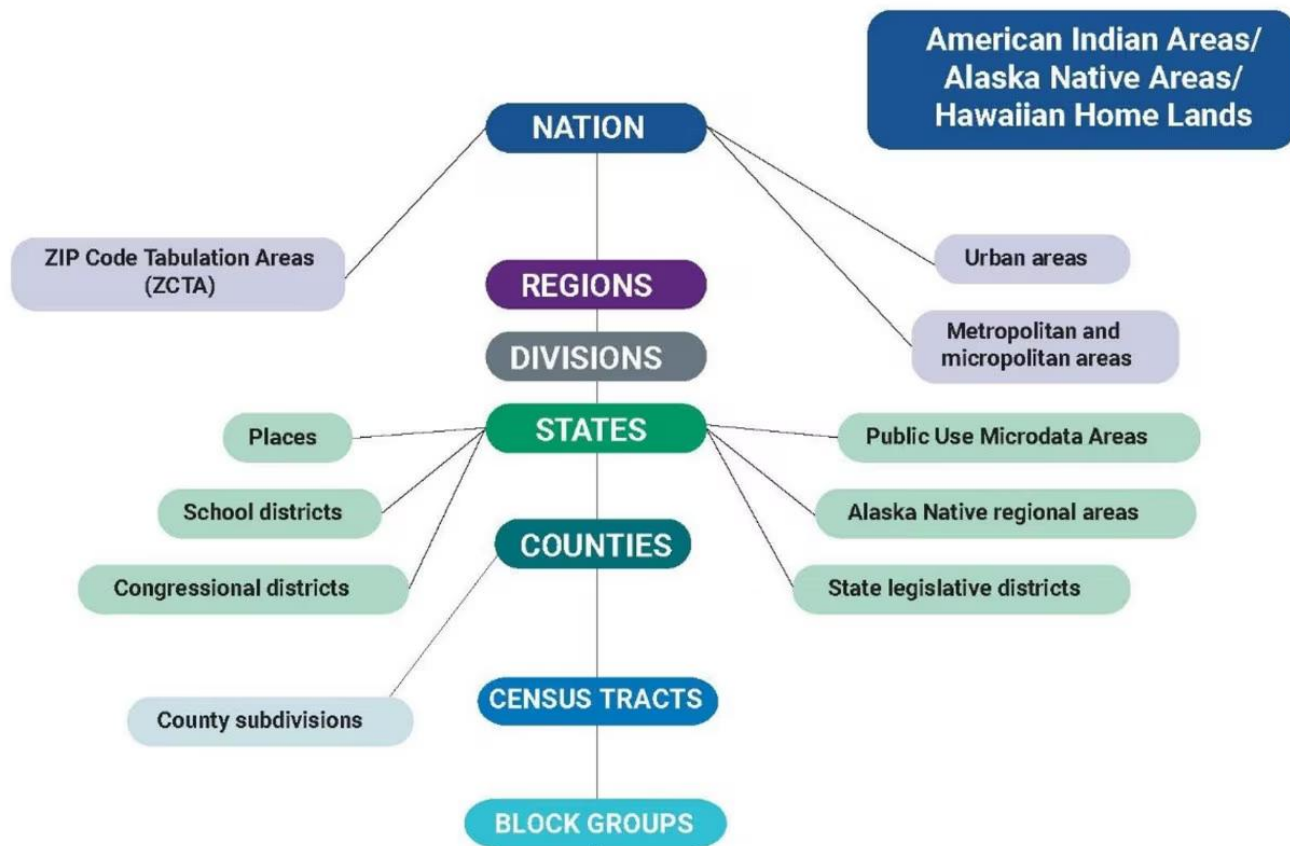
Updated Voter Registration, Turnout, Partisanship (2024, Census Tract)

National Housing Preservation Database

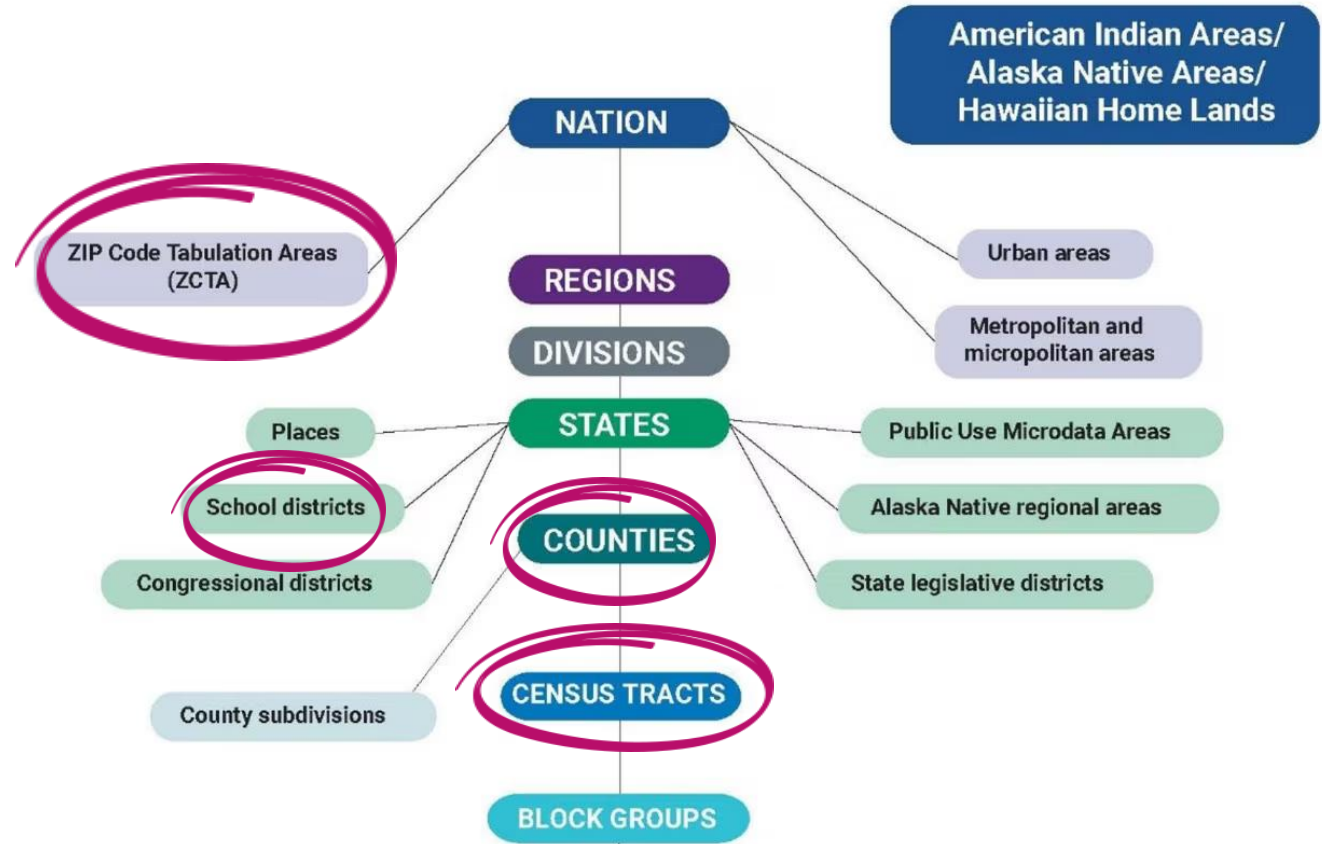
LGBTQ+ Health & Policy

Policing

Hierarchy of select geographic entities



NaNDA geographies



Which geography do I use?

IF YOUR DATA HAVE...

A census tract, FIPS code, or GEOID (11 digits)



JOIN TO NANDA USING...

The FIPS code directly. Check whether it uses 2010 or 2020 geographic boundaries: `tract_fips10` and `tract_fips20` are not interchangeable

A ZIP code



A ZIP-code-to-ZCTA crosswalk. Some ZIP codes have no ZCTA

Full street addresses (not PO boxes)



Geocode first, in ArcGIS or an online geocoder, then join on the resulting FIPS code

“Pre-linked” NaNDA

DATA SOURCE	POPULATION	GEOGRAPHY	PROTOCOL FOR ACCESS
Health and Retirement Study	US adults age 51+	Census tract	Accessed through virtual enclave with restricted data application
National Health and Aging Trends Study	US adults age 65+	Census tract	Accessed through virtual enclave with restricted data application
Panel Study of Income Dynamics	1968 US sample	Block group	Accessed through virtual enclave with restricted data application
Michigan Medicine Electronic Health Records	Michigan Medicine patients	Residential address	IRB approval and U-M affiliation, accessed through DataDirect
Federal Statistical Research Data Centers	Varies by dataset (Census, NCHS, AHRQ surveys)	Census block	Approved project proposal and background check, accessed in a secure research data center

Your data plus NaNDA data, joined on one geographic identifier.



**Your
Data**



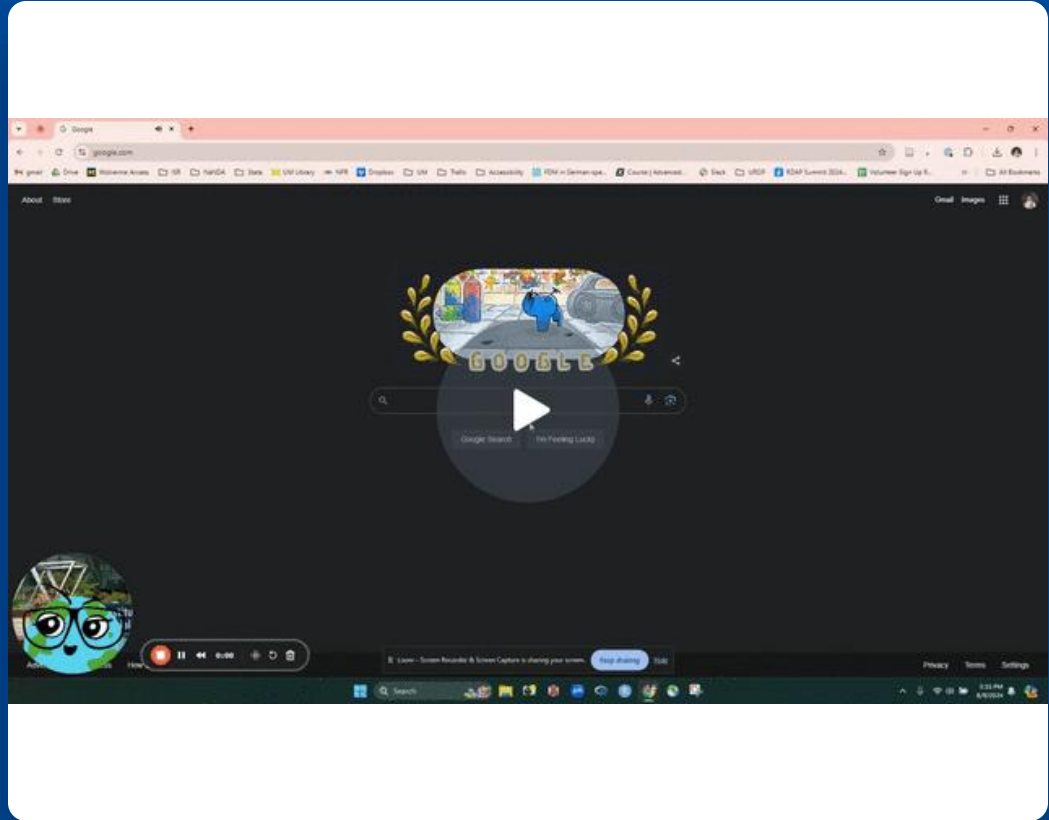
**NaNDA
Data**



New Ideas

NaNDA repository walk through

[Navigating the National
Neighborhood Data Archive](#) 📖
[Watch Video](#)





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Choose your breakout

Back together at 1:45

BEGINNER

Choosing Your First NaNDA Dataset

Pick this if

You are new to neighborhood data, or want help finding the right dataset.

You'll leave with

A NaNDA dataset matched to your geography and years.

INTERMEDIATE

Linking NaNDA with Your Data

Pick this if

Your data have addresses, ZIP codes, or census tracts to merge with NaNDA.

You'll leave with

A merge you ran on practice data, plus code you can reuse.

How to join

1

Click **Breakout Rooms** in your Zoom toolbar. It may be under **More**.

2

Choose **Beginner** or **Intermediate**, then click **Join**.

Not sure? Start with Beginner. You can switch rooms any time.